LYU JIAXIN

Guangzhou, China | 18224734143@163.com | +86 18224734143
linkedin.com/in/jiaxin-lyu81 | github.com/jiaxinlyu-a11y | jiaxinlyu-a11y.github.io

SUMMARY

MSc Statistical Data Science student (UCD) seeking data analyst roles. Analyses and models data in R and Python: regression modelling, sensitivity analysis, simulation and machine learning, with training in predictive analytics. Previously worked with business users on requirements and market analysis in a product role.

EDUCATION

University College Dublin (UCD), MSc Statistical Data Science (GPA 3.54/4.0) *Sep 2025 – Present*

Courses: Machine Learning, Advanced Predictive Analytics, Data Programming with R, Survival Models, Mathematical Statistics, MCMC

Harbin University of Commerce, BSc Statistics *Sep 2019 – Jun 2023*

SKILLS

Analysis: regression and model selection (AIC), hypothesis testing (Cox, log-rank, RMST), sensitivity analysis and diagnostics, simulation, machine learning (CNN, DNN)

Programming: R, Python (OpenCV, Streamlit), SQL, Excel

Reporting & business: Quarto, Markdown, Streamlit; requirements and technical documentation; competitor benchmarking

Tools: Git/GitHub; AI-assisted development (I set requirements and validate)

Languages: English (IELTS 6.5 overall), Mandarin (native)

PROJECTS

KETO-CTA Re-Analysis **| MSc thesis, sole analyst | R, Python**

- **Goal:** using the released data of a retracted imaging study (100 participants, 2 CT scans), test whether baseline plaque and ApoB predict later plaque.
- **My role:** sole analyst. Rebuilt the published figures to check the data reproduced them; as predictors overlap, used joint models and compared 3 regression models (AIC, backward elimination); chose codings from each variable's distribution (log1p, sqrt, binary); ran sensitivity analyses on two outcome definitions and diagnostics (HC3, robust regression).
- **Data recovery (AI-assisted tool):** participant-level ApoB was never released. I set the requirements, had AI write the Python tool (OpenCV, Streamlit) and tested it against known values: on 2 plots of 100 points, 100 and 99 were matched (94 and 87 found automatically, the rest added by hand); mean error 0.13–0.25 mm^3, largest 1.07 mm^3 on an axis of about 160 mm^3 (about 0.7%).

EHS Standard Product Benchmarking **| Work project, product role | competitor analysis, requirements**

- **Goal:** the existing EHS system trailed the market on safety-management flows, ease of use and scenarios; decide what a standard product should do.
- **My role:** reviewed 5 vendors and built a list of about 196 core functions, marking where our system matched, needed work or had nothing; compared 3 working systems function by function (137 functions: done, to improve, not covered); turned the findings into design decisions (plan and task hierarchy, one management page per record type) and adapted the design to one customer's processes.
- **Result:** gap list for the 196 functions: 57 on par, 76 to improve, 63 missing.

Check-up Planner **| AI-assisted tool | JavaScript, bilingual, runs in the browser**

- **Goal:** turn published screening guidelines into check-up advice that adapts to the person, with evidence level and source for each item.
- **My role:** selected 41 rules and fixed their sources (USPSTF, ACS, ACG, Chinese guidelines, OSHA, GBZ 188-2025); designed adjustments for lifestyle, personal and family history where guidelines are silent; tested repeatedly; had AI write the code.
- **Result:** five evidence levels; 9 tests guidelines advise against for people without symptoms are listed separately.

Survival Analysis Simulation | Personal project | R, Quarto. Designed, coded and reported on my own: 24 scenarios × 1,000 repetitions comparing Cox, log-rank and RMST tests; verified Type I error (0.043–0.066 against a 0.05 target) and bias first.

Human Activity Recognition | UCD coursework | R, Keras. Compared a DNN and a CNN on wearable-sensor time series (19 activity classes); the CNN generalised better, with 97.8% test accuracy.

EXPERIENCE

Chengdu VisionVS Technology Co., Ltd., Product Assistant & Delivery Engineer, Chengdu, China *Jul 2023 – Mar 2025*

- Wrote requirements, design specifications and technical documentation with clients for interactive electronic technical manuals (IETM, IETP) and an EHS management system.
- Led an internal project-management system from a survey of four tools to delivery.
- Trained users and worked with developers on iterations.